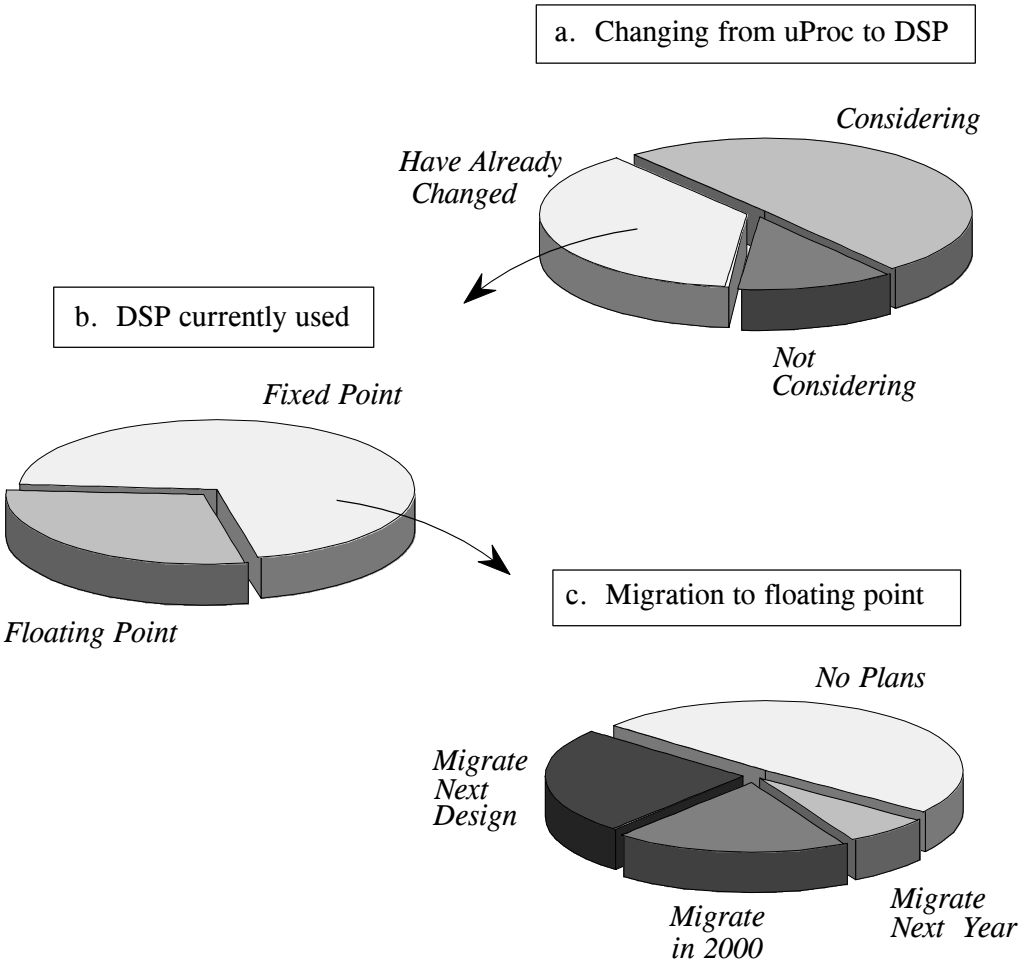

FIGURE 28-8
Major trends in DSPs. As illustrated in (a), about 38% of embedded designers have already switched from conventional microprocessors to DSPs, and another 49% are considering the change. In (b), about twice as many engineers use fixed point as use floating point DSPs. This is mainly driven by consumer products that must have low cost electronics, such as cellular telephones. However, as shown in (c), floating point is the fastest growing segment; over one-half of engineers currently using 16 bit devices plan to migrate to floating point DSPs

instance, suppose you are designing a medical imaging system, such a computed tomography scanner. Only a few hundred of the model will ever be sold, at a price of several hundred-thousand dollars each. For this application, the cost of the DSP is insignificant, but the performance is critical. In spite of the larger number of fixed point DSPs being used, the floating point market is the fastest growing segment. As shown in (c), over one-half of engineers using 16-bits devices plan to migrate to floating point at some time in the near future.

Before leaving this topic, we should reemphasize that floating point and fixed point usually use 32 bits and 16 bits, respectively, *but not always*. For